COMPREHENSIVE TECHNICAL SPECIFICATION NOTE

Deep-Dive Weighted Risk Scoring Engine (WRSE) & Dynamic Master Collision Architecture
ContextGuard Framework | Enterprise Healthcare & TechMed Shadow AI Governance

1. Architectural Foundation & Multi-Factor Mathematical Formulation

The ContextGuard framework provides a low-latency, deterministic governance layer for enterprise Shadow AI interactions. ContextGuard dynamically evaluates data sensitivity (S), target AI destination trust (D), and actor authority (U) in real-time.

$$RS = (W_s * S) + (W_d * D) + (W_u * U)$$

- **W_s = 0.50:** Data Sensitivity Weight (primary factor reflecting corporate asset sensitivity).
- **W_d = 0.25:** Destination AI Trust Weight (reflecting target provider governance and security guarantees).
- **W_u = 0.25:** User Identity and Authority Weight (reflecting workstation role and exfiltration blast radius).

2. Exhaustive Registry Taxonomy & Attribute Schema (All 18 Datasets)

ContextGuard operates across 18 enterprise asset registries structured into three security tiers:

Tier & Domain	CSV Registry Filename	Deterministic High-Entropy Keys	Quasi-Identifiers & Low-Entropy Fields	Base S
Tier 1: PHI	T1_ClinicalDiagnosis.csv	RECORD ID (CLN-XXXXXX), PATIENT ID (TM-202X-XXXX), ICD 10 CODE	PATIENT NAME, PRIMARY DIAGNOSIS, ATTENDING PHYSICIAN, DEPARTMENT, CLINICAL SUMMARY, DATES, SEVERITY	**0.95**
Tier 1: PHI	T1_ClinicalDiagnosis_registry (1).csv	RECORD ID (CLN-XXXXXX), PATIENT ID (TM-202X-XXXX), ICD 10 CODE	PATIENT NAME, DIAGNOSES, ATTENDING PHYSICIAN, DEPARTMENT, CLINICAL SUMMARY, SEVERITY	**0.95**
Tier 1: PHI	T1_ClinicalDiagnosis_registry.csv	RECORD ID (CLN-XXXXXX), PATIENT ID (TM-202X-XXXX), ICD 10 CODE	PATIENT NAME, DIAGNOSES, ATTENDING PHYSICIAN, DEPARTMENT, CLINICAL SUMMARY, SEVERITY	**0.95**
Tier 1: PHI	T1_Insurance.csv	RECORD ID (INS-XXXXXX), PATIENT ID, INSURANCE ID, POLICY NUMBER, MEDICARE ID, CLAIM NUMBER	PATIENT NAME, INSURANCE PROVIDER, PAYER IDENTITY, CLAIM AMOUNTS, STATUS, CLAIM DATE	**0.95**
Tier 1: PHI	T1_LabTelemetry.csv	RECORD ID (LAB-XXXXXX), PATIENT ID, DNA SEQUENCE LOG, DICOM/MRI REPORT ID	PATIENT NAME, TEST TYPE, HEMOGLOBIN, WBC COUNT, PLATELETS, GLUCOSE, CREATININE, LAB TECH	**0.95**
Tier 1: PHI	T1_PatientCoreIDs.csv	RECORD ID (REC-XXXXXX), PATIENT ID, SSN, PASSPORT NUMBER, EMERGENCY CONTACT	PATIENT NAME, DOB, RESIDENTIAL ADDRESS, GENDER, BLOOD TYPE,	**0.95**

		TEL	NATIONALITY, TIMESTAMPS	
Tier 1: PHI	T1_Pharmacy.csv	RECORD ID (RX-XXXXXX), PATIENT ID, PRESCRIPTION MATRIX ID, PHARMACEUTICAL LOG REF	PATIENT NAME, MEDICATION LIST, DOSAGE, DRUG ALLERGY, PRESCRIBING DOCTOR, BRANCH, REFILLS	**0.95**
Tier 2: Infra	T2_DBConnections.csv	RECORD ID (DB-XXXXXX), DB SERVER IP, DB CONNECTION STRING, BACKUP CLUSTER IP	ASSET TYPE, DB TYPE, DATABASE NAME, POOL SIZE, DB VERSION, ENCRYPTION, ENVIRONMENT	**0.90**
Tier 2: Infra	T2_IAM.csv	RECORD ID (IAM-XXXXXX), SAML TOKEN HASH, KERBEROS REALM, LDAP BASE DN	ASSET TYPE, USERNAME, IAM ROLE, DEPARTMENT, AD FOREST, MFA ENABLED, PRIVILEGE LEVEL	**0.90**
Tier 2: Infra	T2_NetworkNodes.csv	RECORD ID (NET-XXXXXX), IP ADDRESS, BACKUP GATEWAY IP, DOMAIN CONTROLLER FQDN	ASSET TYPE, SERVER NAME, ACTIVE DIRECTORY OU, DNS ZONE, OS VERSION, PATCH LEVEL, LOCATION	**0.90**
Tier 2: Infra	T2_NetworkTopology.csv	RECORD ID (TOP-XXXXXX), SUBNET (CIDR), GATEWAY IP, ROUTING TABLE ENTRY	ASSET TYPE, VLAN TAG, DMZ ZONE, NETWORK SEGMENT, BANDWIDTH MBPS, REDUNDANCY, LOCATION	**0.90**
Tier 2: Infra	T2_SecurityPerimeter.csv	RECORD ID (SEC-XXXXXX), FIREWALL RULE ID, VPN TUNNEL PROFILE, PROXY SERVER IP	ASSET TYPE, NAT POLICY, SSL INSPECTION, IDS/IPS STATUS, VENDOR, PRIORITY, COMPLIANCE	**0.90**
Tier 3: IP	T3_HL7Protocols.csv	RECORD ID (HL7-XXXXXX), DICOM STUDY ID, MLLP PORT, FHIR RESOURCE ENDPOINT	ASSET TYPE, HL7 MESSAGE HEADER, SENDING SYSTEM, RECEIVING SYSTEM, MESSAGE TYPE, ENGINE	**0.85**
Tier 3: IP	T3_LabAlgorithms.csv	RECORD ID (ALG-XXXXXX), CALIBRATION CONSTANT, TELEMETRY ENDPOINT, SCRIPT NAME	ASSET TYPE, ALGORITHM VERSION, ANALYZER MODEL, TEST CATEGORY, ACCURACY PERCENT, LOGIC REF	**0.85**
Tier 3: IP	T3_SourceCodeAPI.csv	RECORD ID (SRC-XXXXXX), API KEY HASH (sk_live_), JWT SECRET ID, API SECRET REF	ASSET TYPE, GIT REPOSITORY, LANGUAGE, MODULE NAME, VERSION, CLOUD PROVIDER, DEPLOY ENV	**0.85**
Tier 3: IP	T3_StrategicBlueprints.csv	RECORD ID (STR-XXXXXX), DOCUMENT TAG, MERGER TARGET CO, TRIAL DRUG CODE	ASSET TYPE, PROJECT NAME, FINANCIAL FORECAST USD M, CLINICAL TRIAL PHASE,	**0.85**

			STRATEGIC OWNER	
Tier 3: IP	T3_StrategicBlueprints_registry.csv	RECORD ID (STR-XXXXXX), DOCUMENT TAG, MERGER TARGET CO, TRIAL DRUG CODE	ASSET TYPE, PROJECT NAME, FINANCIAL FORECAST USD M, CLINICAL TRIAL PHASE, STRATEGIC OWNER	**0.85**
Tier 3: IP	T3_VendorContracts.csv	RECORD ID (VND-XXXXXX), CONTRACT ID, PRICING SHEET REF, NDA TYPE	ASSET TYPE, VENDOR NAME, PROCUREMENT BUDGET USD, CONTRACT DATES, PAYMENT TERMS, SIGNED BY	**0.85**

3. Two-Tier Granular Sensitivity Scoring & Context Correlation

A. Class 1: Standalone High-Entropy Primary Identifiers (Zero-Tolerance)

- **API & Cryptographic Secrets:** Regex sk_live_[a-zA-Z0-9]{20,}, JWT Tokens, SAML Assertion Hashes.
- **Deterministic Record Prefixes:** CLN-, INS-, LAB-, REC-, RX-, DB-, IAM-, NET-, TOP-, SEC-, HL7-, ALG-, SRC-, STR-, VND- followed by 6 numeric digits.
- **Healthcare & Citizen Identifiers:** Patient ID format TM-202[0-9]-[0-9]{4}, SSN, Passport Numbers, Medicare ID.
- **Internal Network & Infrastructure:** Private RFC-1918 IPv4 Blocks (10.x, 172.16-31.x, 192.168.x) and database URIs.

B. Class 2: Quasi-Identifiers & Context Correlation Logic

- **Isolated Occurrence:** Assigned S = 0.00 to prevent false-positive critical escalations on benign queries.
- **Combinatorial Leak:** When >= 2 quasi-identifiers match the same record, Context Stitching elevates S to 0.95.

4. Dynamic Environmental Coefficients (D and U Matrices)

- **Destination Matrix (D):** Public Unmanaged (D=0.95) | Enterprise Managed (D=0.30) | Local On-Premise (D=0.10).
- **User Authority Matrix (U):** Privileged Admin (U=0.90) | Medical Specialist (U=0.75) | Standard Staff (U=0.65) | Automated Bot (U=0.20).

5. Ingestion Pipeline & Execution Steps

- **Step 1 (Payload Interception):** Capture HTTP POST via mitmproxy and decode entity wrappers (e.g., Google Gemini f.req=).
- **Step 2 (In-Memory Hash Collision):** Build in-memory hash indices of all 18 CSV datasets for O(1) matching.
- **Step 3 (Identifier Classification):** Sweep for Class 1 tokens (assign S=0.95/0.90/0.85) or correlate Class 2 attributes.
- **Step 4 (WRSE Aggregation):** Calculate RS = (0.50 * S) + (0.25 * D) + (0.25 * U) scaled to [0% - 100%].

- **Step 5 (Severity Tagging & Persistence):** Map to CRITICAL (>80%), HIGH (70-80%), MEDIUM (55-70%), LOW (<55%) and save to users.db.

6. Verification Traces

Case 1 (PHI Header Leak): U=0.90, D=0.95, S=0.85 (T3_HL7Protocols) -> RS = (0.50*0.85)+(0.25*0.95)+(0.25*0.90) = 88.75% (CRITICAL)

Case 2 (API Key Exfiltration): U=0.65, D=0.95, S=0.85 (T3_SourceCodeAPI) -> RS = (0.50*0.85)+(0.25*0.95)+(0.25*0.65) = 82.50% (CRITICAL)

Case 3 (Benign Query): U=0.65, D=0.95, S=0.00 (Isolated Name) -> RS = (0.50*0.00)+(0.25*0.95)+(0.25*0.65) = 40.00% (LOW)

7. Core Engineering Advantages

- **Zero False Positives:** Contextual correlation prevents common workplace vocabulary from triggering false critical blocks.
- **Deterministic Sub-Millisecond Latency:** Vectorized in-memory hash lookups eliminate secondary LLM inference bottlenecks.